12. MATHEMATICAL OPERATIONS

This section deals with questions on simple mathematical operations. Here, the four fundamental operations — addition, subtraction, multiplication and division and also statements such as less than\ 'greater than', 'equal to', 'not equal to[1], etc. are represented by symbols, different from the usual ones. The questions involving these operations are set using artificial symbols. The candidate has to substitute the real signs and solve the questions accordingly, to get the answer.

TYPE 1 : PROBLEM-SOLVING BY SUBSTITUTION

In this type, you are provided with substitutes for various mathematical symbols, followed by a question involving calculation of an expression or choosing the correct/ incorrect equation. The candidate is required to put in the real signs in the given equation and then solve the questions as required.

Note : While solving a mathematical expression, proceed according to the rule BODMAS — *i.e.,* Brackets, Of, Division, Multiplication, Addition, Subtraction.

eg., (36 -12) + 4 + 6 + 2 x 3 = 24 + 4 + 6 + 2 x 3 (Solving Bracket)
= 6 + 3 x 3 (Solving Division)
= 6 + 9 (Solving Multiplication)
= 15 (Solving Addition)

ILLUSTRATIVE EXAMPLES"

Ex. 1. If V means 'divided by[1], [4]- means 'multiplied by', V means 'minus and V means 'plus', which of the following will be the value of the expression 16+ 8- 4 + 2 x 4 ? (Bank P.O. 1995)

(a) 16 (6) '28 (c) 32 *(d)* 44 (e) None of these

Sol. Putting the proper signs in the given expression, we get :
16+ 8 x 4 + 2- 4 = 16+ 8 x 2 - 4 = 16+ 16 - 4 = 32 - 4 = 28.
So, the answer is (6).

Ex. 2. If + means +, - means x, + means + and x means then
36 x 12 + 4 + 6 + 2 - 3 = ?

(a) 2 (6) 18 (c) 42 *Id)* 6 | (e) None of these

Sol. Using the proper signs, we get :
36- 12 + 4 + 6 + 2 x 3 = 36 - 3 + 3 x 3 = 36 - 3 + 9 = 45 - 3 = 42.
So, the answer is (c).

Ex. 3. If A means 'plus', B means 'minus', C means 'divided by' and D means 'multiplied by', then 18 A 12 C 6 D 2 B 5 = ? (B.S.R.B. 1996)

(a) 15 (6) 25 (c) 27 (tf) 45 *(e)* None of these

Sol. Using the proper signs, we get :
Given expression = 18+12+ 6 x 2 - 5 = 18+ 2 x 2 - 5
= 18 + 4- 5 - 2 2 - 5 = 17.
So. the answer is (*).

Ex. 4. If x stands for + stands for +, + stands for + and - stands for x, which one of the following equations is correct ? (S.8.C. 199«)

(a) 15 - 5 + 6 x 20 + 10 = 6 (6) 8 + 10 - 3 + 5 x 6 = 8

(c) 6 x 2 + 3 + 12 - 3 = 15 *(d)* 3 + 7 - 5 x 1 0 + 3 = 10

8oL Using the proper signs, we get :

Expression in (a) = 15 x 5 + 5 - 20 + 10 = 15 x 5 + 5 - 2 = 75 + 5 - 2 = 78.

3 "

Expression in(6)=8+10x3 + 5- 6 = 8 + 1 0 x - - 6 = 8 + 6- 6 = 8.

2 2 124

Expression in (c) = 6 - 2 + 3 + 12x3 = 6 - 1 + 36 = 4 2 - 1 = ^-

I V. u *o o*

Expression in *(d)* = 3 + 7x5 - 10 + 3 = 3 + 7 x 5 - ^ = 3^ 35 - ^ = ^ •

O O u

Statement (b) is true.

Ex. 5. It being given that : > denotes $+_f$ < denotes $-_f$ + denotes $+_t$ - denotes =, = denotes less than' and x denotes 'greater than', find which of the following is a correct statement.

(a)3 + 2 > 4 « 9 + 3 < 2 (6) 3 > 2 > 4 = 18 + 3 < 1

(c)3 > 2 < 4 x 8 + 4 < 2 (d)3 + 2 < 4 x 9 + 3 < 3

Sol. Using proper notations, we have :

(a) Given statement is3 + 2 + 4 < 9 + 3- 2 or — < 1, which is not true.

(6) Given statement is 3 + 2 + 4< 18 + 3 - 1 or 9<5, which is not true.

(c) Given statement is3 + 2- 4 > 8 + 4- 2 or 1>0, which is true.

(d) Given statement is3 + 2- 4 > 9 + 3- 3 or -1 > 0, which is not true.

So, the statement (c) is true.

EXERCISE 12A

1. If x stands for 'addition', + stands for 'subtraction', + stands for 'multiplication' and - stands for 'division', then

20x8 + 8- 4 + 2- ? (Transmission Executives' 1994)

(a) 80 (6) 25 (c) 24 *id)* 5

2. If - means x_f x means +, + means 4- and + means then

40x 12 + 3- 6 + 60 = ? (Bank P.O. 1993)

(a) 7.95 (6) 16 (c) 44 *(d)* 479.95 (e) None of these

3. If + means +, x means + means x and - means +, then

8 + 6 * 4 + 3- 4 * ? (Bank P.O. 1994)

(a) - 12 (6) - ~9ft ' <c> 12 <rf> 9ft y <*> None of these

4. If x means +, - means x, + means + and + means then

(3 - 15 + 19) x 8 -f 6 = ? (Assistant Grade, 1998)

(a) 8 (6)4 (c) 2 *(d)* - 1

5. If + means x, + means x means + and - means +, what will be the value of 4 + 11 + 5 - 5 5 = ? (LAC. 1994)

(a) - 48.5 (6) - 11 (c) 79 *(d) 91* *(e)* None of these

6. If * means * means -, - means x and + means +, then
8 x 7 - 8 * 4 0 + 2 = ? (Bank P.O. 1998)

(a) 1 (6) 7 ? (c) 81 (d) 44 (e) None of these
5 5

7. If + means - means x, x means + and + means +, then
1 5 x 3 + 1 5 * 5 - 2 - ? (S.B.IJP.O. 1994)
(a) 0 (6)6 (c) 10 W) 20 (e) None of these
8. If x means + means +, - means x and + means +, then
15 - 2 + 900 + 90 * 100 = ? (B.S.R.B. 1996)
(a) 19Q (6) 180 (c) 90 (d) 0 (e) None of these
9. \(+ means +, - means x, + means -, x means +. what will be the value of 8 + 6 + 4 - 7 x 3 ?
(a) - ^ (6) - ~ (c) 12 *id)* 14 (e) None of these
o /
10. If + means +. - means +, x means - and + means x, then
(36 K 4) - 8 X 4 ,
4 + 8 x 2 + 1 6 + 1 "
(a) 0 (6) 8 (c) 12 (</) 16
11. If P denotes +, Q denotes x, R denotes - and S denotes then
18 Q 1 2 P 4 R 5 S 6 = ?
(a) 36 (6) 53 (c) 59 (d) 65 (e) None of these
12. If a means 'plus', *b* means 'minus', c means 'multiplied by' and *d* means 'divided by', then 18 *c* 14 *a* 6 6 16 *d* 4 = ? (B.S.R.B. 1996)
(a) 63 (6) 254 (c) 288 (cf) 1208 *(e)* None of these
13. If A means -, B means +, C means + and D means x, then
15 B 3 C 24 A 12 D 2 = ? (Bank P.O. 1996)

(a) 34 (6) 2 (c) | (rf) - 2 3 1 *ie)* None of these

14. If * stands for 'add', *y* stands for 'subtract', *z* stands for 'divide' and *p* stands for 'multiply*, then what is the value of (7 *p* 3) y 6 *x* 5 ? (UJ).C. 1994)
(a) 5 (6) 10 \ (c) 15 *(d)* 20
15. If A stands for +, B stands, for C stands for x, then what is the value of (10 C 4) A (4 C 4)B 6 ? (Assistant Grade, 1992)
(a) 60 (6) 56 (c) 50 (d) 46
16. If L denotes x, M denotes +, P denotes + and Q denotes -, then
16 P 24 M 8 Q 6 M 2 L 3 = ?

. (a) ~ (6) (c) 14 £ (<f) 10 (e) None of these
b b T

17. If - means + means x, + means * means +, then which of the following equations is correct ? (C.B.I. 1997)
(a) 52 + 4 + 5 x 8 - 2 = 36 • (6)4 3x7 + 5 + 4- 8 = 25
(c) 36 x 4 - 12 + 5 + 3 « 420 *(d)* 3 6 - 1 2 x 6 + 3 + 4 = 60
18. If x means 'addition', - means 'division', + means "subtraction* and means 'multiplication', then which of the following equations is correct ? (8.8.C. 1996)

(a) 16 x 5 + 10 + 4 - 3 = 19 (6) 16 + 5 + 10 x 4 - 3 = 9
(c) 16 + 5 - 10 x 4 + 3 * 9 ' (d) 16 - 5 x 10 + 4 + 3 = 12

19. If + stands for 'division', x stands for 'addition', - stands for 'multiplication' and + stands for 'subtraction', then which of the following equations is correct ?
(a) 36 x 6 • 7 + 2 - 6 = 20 (6) 36 + 6 + 3 x 5 - 3 = 45
(c) 36 + 6 - 3 x 5 + 3 = 24 (d) 36 - 6 + 3 x 5 + 3 = 74
(Assistant Grade, 1994)

20. If P denotes +, Q denotes R denotes x and S denotes +, which of the following statements is correct ?
(a) 36 R 4 S 8 Q 7 P4 = 10 (6) 16 R 12 P49S 7 Q9 = 200
(c) 32 S 8 R 9 = 160 Q 12 R 12 (d) 8 R 8 P 8 S 8 Q 8 = 57

21. If L denotes +, M denotes x, P denotes t and Q denotes then which of the following statements is true ?
(a) 32 P 8 L 16 Q 4 = - 1 Z (6) 6 M 18 Q 26 L 13 P 7 = -J.3 ,l lu
(c) 11 M34L 17Q8L3 = ^ (d) 9 P9 L9 Q 9 M 9 = - 71

22. If x stands for 'addition', < for 'subtraction', + stands for 'division', > for 'multiplication'. - stands for 'equal to'. + for 'greater than' and «= stands for 'less than', state which of the following is true ? (U.D.C. 1994)
(a)3x2<4 + 16>2 + 4 (6)5>2 + 2 = 10<4x8
(c) 3 x 4 > 2 - 9 + 3 < 3 (d) 5x3 <7 + 8 + 4 x 1

Directions *(Questions 23 to 27)* : ***If > denotes +, < denotes -, + denotes +, X denotes x, - denotes =, x denotes > and = denotes <, choose the correct statement in each of the following questions.***

23. (a)6 + 3>8 = 4 + 2<l (6)4>6 + 2x32 + 4<l
(c)8<4 + 2 = 6>3 (d) 14 + 7 > 3 = 6 + 3 > '2
24. (a) 14 > 18 + 9 = 16 + 4 < 1 (6) 4 > 3 A 8 < 1 - 6 + 2 > 24
(C)3<6A4>25 = 8 + 4>1 (d)12>9 + 3<6x25 + 5>6
25. (a) 13 > 7 < 6 +£ = 3 A 4 (6) 9 > 5 > 4 - 18 + 9 > 16
(C)9<3<2>1X8A2 (d) 28 + 4A2 = 6A4 + 2
26. (a) 29 < 18 + 6 = 36 + 6 A 4 (6) 18 > 12 + 4 x 7 > 8 A 2
(c) 32>6 + 2 = 6<7A2 (d)31>l<2 = 4>6A7
27. (a) 7 > 7 < 7 + 7 = 14 (6)7A7>7 + 7 = 7A7>1
(c) 7 < 7 + 7 = 6 (d)7-7>7 = 8

Directions *(Questions 28 to 32)* : ***In each of the following questions, different alphabets stand for various symbols as indicated below :***

Addition : O ***Subtraction : M*** ***Multiplication : A***
Division : Q ***Equal to : X*** ***Greater than : Y***
Less than : Z (I. Tax ***A*** Central Excise, 1996)

Out of the four alternatives given in these questions, only one is correct according to the above letter symbols. Identify the correct answer.

28. (a) 2 Z 2 A 4 0 1 A 4 M 8 (6) 8 Y 2 A 3 A 4 Q 2 A 4
(c) 10 X 2 O 2 A 4 O 1 M 2 (d) 12 X 4 O 2 Q 1 A 4 A 2

29. (a) 1 O 1 Q 1 M 1 Y 3 Q 1 (6) 2 Q 1 O 10 A 1 Z 6 A 4
(c) 3 O 2 O 10 Q 2 X 10 A 2 (d) 5 Q 5 A 5 0 5 Y 5 A 2
30. (a) 3 O 2 X 2 Q 1 A 3 O 1 (6) 6 M 2 Y 10 Q 2 A 3 O 1
(c) 10 A 2 Z 2 Q 2 A 10 Q 2 (d) 10 A 2 Y 2 Q 1 A 10 Q 2
31. (a) 32 X 8 Q 2 A 3 Q 1 A 2 (6) 14 X 2 A 4 A 2 M 2 Q 1
(c) 2 Y 1 A 1 Q 1 O 1 A 1 (d) 16 Y 8 A 3 O 1 A 2 M 2
32. (a) 8 Q 4 A 1 M 2 X 16 M 16 (6) 8 O 2 A 12 Q 10 X 18 Q 9
(c) 6 Q 2 O 1 O 1 X 16 A 1 (< f) 2 0 3 M 4 Q 2 Z I A (2

Directions (Questions 33 *to 37*): *In the following questions, different letters stand for carious symbols as indicated below :*

R : Addition ***S : Subtraction*** ***T: Multiplication***
U: Division ***V: Equal to*** ***W : Greater than***
X: Less than

Out of the four alternatives given in these questions, only one is correct according to the above letter symbols. Identify the correct one.

33. (a) 16 T 2 R 4 U 6 X 8 (6) 16 R 2 S 4 V 6 R 8 °
(c) 16 T 2 U 4 V 6 R 8 (d) 16 U 2 R 4 S 6 W 8
34. (a) 20 U 4 R 4 X 2 T 3 (6) 20 S 4 U 4 V 2 T 3
(c) 20 T 4 U 4 U 2 X 3 (d) 20 R 4 U 4 S 2 W 3
35. (a) 15 U 5 R 3 V 2 T 3 (6) 15 U 5 W 3 R 2 T 3
(c) 15 S 5 T 3 W 2 R 3 (<f) 15 R 5 U 3 V 2 R 3
36. (a) 24 U 3 R 2 S 2 W 8 ***(b)*** 24 S 3 X 2 T 2 U 8
(c) 24 R 3 S 2 X 2 T. 8 " *id)* 24 U 3 T 2 V 2 T 8
37. (a) 30 R 6 U 2 W 4 T 3 (6) 30 S 6 S 2 X 4 T 3
(c) 30 S 6 U 2 U 4 V 3 (<f) 30 U 6 R 2 W 4 T 3

ANSWERS

1.(c): Using the correct symbols, we have :
Given expression =20+ 8- 8 + 4x2
• 20 + 8 - 2x2 - 20 + 8 - 4 = 24.

2. (e): Using the correct symbols, we have :
Given expression =40+12 + 3x6-60
= 40 + 4 x 6 -60 = 40 + 24-60-4.

3. (6) Using the correct symbols, we have :
Given expression - 8 + 6- 4x3 + 4

4. (c): Using the correct symbols, we have :
Given expression = (3 x 15 + 19) + 8 - 6
«(45 + 19) + 8- 6 = 64 + 8- 6«=8-6 = 2. ^

5. (e): Using the correct symbols, we have :
Given expression « 4 x 11 - 5 + 55 = 44 - 5 + 55 = 94.

6. (6): Using the correct symbols, wo have
Given expression = 8 + 7x8 +40- 2

= 8 + 7 x i - 2 = 8 + 1 - 2 = ^ = 7^
o 0 5 5

7. (c): Using the correct symbols, we have :
Given expression = 15 + 3+ 15- 5x2 = 5 + 1 5 - 5 x 2 x 5 + 15- 10= 10.

8. ie): Using the correct symbols, we have :
Given expression = 15 x 2 + 900 + 90 - 100
= 15 x 2 + 10 - 100 = 30 + 10 - 100 = - 60

9. (a): Using the correct symbols, wc have :
Given expression »8 f 6 - 4 x 7 ^ 3
« J - 4 x 7 + 3 * | - 2 8 * 3 = - y .

10. *(a)* : Using the correct symbols, we have :
Given expression-^8-2x16+^1 (36 - 4)+ 8 -4
32 + 8 - 4   4 -4   n
" 3 2 - 3 2 + 1 a i ~ = 0

11. *ib)*: Using the correct symbols, we have :
Given expression « 18 * 12 + 4 + 5- 6
= 18 * 3 + 5 - 6 = 54 + 5 - 6 = 53.

12. (h): Using the correct symbols, we have :
Given expression = 18 x 14 + 6 - 16 + 4
= 18 x 14 + 6 - 4 = 252 • 6 - 4 - 254.

13. *ie)*: Using the correct symbols, we have :
Given expression * 15 + 3 + 24 - 12 x 2
r 5 * 2 4 - 1 2 * 2 = 5 + 24-24 = 5

14. *id)*: Using the corrcct symbols, we have :
Given expression =* (7 x 3) - 6 + 5 = 21 - 6 • 5 • 20

15. (c): Using the correct symbols, we have :
Given expression = (10 x 4) + (4 x 4) - 6 = 40 + 16 - 6 » 50.

16. *id)*: Using the correct symbols, we have :
Given expression =16 + 24+ 8- 6 + 2x3
= 16 + 3 - 3 x 3 = 16 + 3 - 9 = 10.

17. *ia)* : Using the proper notations in (a), we get the statement as
52- 4x5 + 8 + 2 = 52- 4*5 + 4= 52-20- 4 = 36

18. (c): Using the proper notations in (c), we get the statement as
16*5 + 10 + 4- 3 * 1 6 * 1 + 4- 3 = 8 + 4- 3 = 9.

19. (rf): Using the proper notations in (d). we get the statement as
36x6 + 3* 5 - 3 = 36 x 2 + 5 - 3 * 72 + 5 - 3 - 74.

20. *id):* Using the proper notations in (d), we get the statement as
8 x 8 * 8 + 8 - 8 = 8- 8 + 1 - 8 = 64 + 1- 8* 57.

21. *id):* Using the proper notations in (d), we get the statement as
9 t 9 r 9 - 9 * 9 = 9+ I- 9x9 = 9 * I - 8 1 = - 71.

22. *lb):* Using the proper notations in *ib).* we get the statement as
5 x 2 2 < 10 •• 4 + 8 or 5 < 14. which is true. ^

23. (c): Using the proper notations in (eft* we get the statement as
8 - 4 + 2<6 + 3 or 6<9. which is true.

24. *(b)* : Using the proper notations in (fe), we get the statement as
4 + 3 x 8 - 1 = 6 + 2 +24 or 27 = 27. which is true.

25. (6): Using the proper notations in (6), we get the statement as
9 + 5 + 4 = 18 + 9 + 16 or 18 = 18. which is true.
26. (d) : Using the proper notations in (d). we get the statement as
3 1 + l - 2 < 4 + 6 x 7 or 30<46. which is true.
27. (a): Using the proper notations in (a), we get the statement as
7 + 7 - 7 + 7 < 14 or 13 < 14, which is true.
28. (a): Using the proper notations in (a), we get the statement as
2 < 2 x 4 + l x 4 - 8 or 2 <4, which is true.
29. (6) : Using the proper notations in (fc), we get the statement as
2 + 1 + 10 x 1 < 6 x 4 or 12 < 24, which is true.
30. (d) : Using the proper notations in (d), we get the statement as
10 x 2 > 2 + 1 x 10 + 2 or 20 > 10, which is true.
31. (6) : Using the proper notations in (6), we get the statement as
14 = 2 x 4 x 2 - 2 + 1 or 14 = 14, which is true.
32. (a): Using the proper notations in (a), we get the statement as
8 + 4 x 1 - 2 = 16-16 or 0 = 0, which is true.
33. (6): Using the proper notations in (6), we get the statement as
16+ 2- 4 = 6 + 8 or 14 = 14, which is true.
34. (d) : Using the proper notations in (d). we get the statement as
20 + 4 + 4 - 2 > 3 or 19 > 3, which is true.
35. *(a)*: Using the proper notations in (a), we get the statement as
15 + 5 + 3 = 2 x 3 or 6= 6, which is true.
36. (d) : Using the proper notations in (d), we get the statement as
24 + 3 x 2 = 2 x 8 or 16 = 16, which is true.
37. (a): Using the proper notations in (a), we get the statement as
30 + 6 • 2 > 4 x 3 or 33 > 12, which is true.

TYPE 2: INTERCHANGE OF SIGNS AND NUMBERS

Ex. 1. If the given interchanges namely : signs + and + and numbers 2 and 4 are made in signs and numbers, which one of the following four equations would be correct ?

(a) 2 + 4 + 3 = 3 (6) 4 + 2 + 6 = 1.5 (c) 4 + 2 + 3 = 4 (rf) 2 + 4 + 6 = 8

Sol. Interchanging + and + and 2 and 4, we get :

(a) 4 + 2 + 3 = 3 or 5 = 3, which is false.

(б) 2 + 4 + 6 = 1.5 or 6.5 = 1.5, which is false.

(c) 2 + 4 + 3 = 4 or ^ = 4, which is false.
o '

(d) 4 + 2 + 6 = 8 or 8 = 8, which is true.

Ex. 2. Which one of the four interchanges in signs and numbers would make the given equation correct ?

3 + 5 - 2 = 4

(a) + and 2 and 3 (6) + and 2 and 5

(c) + and 3 and 5 (d) None of these

Sol. By making the interchanges given in (a), we get the equation as
2 - 5 + 3 = 4 or 0 = 4, which is false.

By making the interchanges given in (6), we get the equation as
3 - 2 + 5 = 4 or 6 - 4 , which is false.
By making the interchanges given in (c), we get the equation as
5 - 3 + 2 = 4 or 4 = 4, which is true.
So, the answer is (c).

EXERCISE 126

Directions *(Questions 1 to 4) : In each of the following questions if the given interchanges are made in signs and numbers, which one of the four equations would be correct f*

1. Given interchanges : Signs - and * and numbers 4 and 8.-
   (a) 6 - 8 + 4 = - I . (6) 8 - 6 + 4 = 1
   (c) 4 -K8 - 2 = 6 (d) 4 - 8 + 6 = 2
2. Given interchanges Signs + and x and numbers 4 and 5.
   (a) 5 x 4 + 20 = 40 (6) 5 x 4 + 20 = 85
   (c) 5 x 4 + 20 = 104 (d) 5 x 4 + 20 = 95
3. Given interchanges Signs + and - and numbers 4 and 8.
   (a[1]) 4 + 8 - 1 2 = 16 (6) 4 - 8 + 1 2 = 0
   8 + 4 - 12 = 24 <d) 8 - 4 + 12 = 8
4. Given interchanges Signs - and x and numbers 3 and 6.
   (a) 6 - 3 x 2 = 9 (6) 3 - 6 i 8 • 10
   (c) 6 x 3 - 4 = 15 (d) 3 x 6 - 4 = 33
5. Find out the two signs to be interchanged for making following equation correct:
   5 + 3 x 8 - 1 2 + 4 = 3 (CJLT. 1997)
   (a) + and - " (6) - and + (c) + and x (d) + and +

Directions *(Questions 6 to 10) : In each of the following questions, an equation becomes incorrect due to the interchange of two signs. One of the four alternatives under it specifies the Interchange of signs in the equation* which when made will make the equation correct. Find the correct alternative.*
(U.D.C. 1991)

6. 5 + 6 + 3 - 12 x 2 « 17
   (a) • and x (6) + and x *(c)* + and + (d) + and -
7. 2 x 3 + 6 - 1 2 + 4 » 17
   (a) x and + ***(b)*** + and - (c) + and + (d) - and +
8. 16 8 + 4 + 5 x 2 « 8
   (a) + and x (6) - and + (c) + and + (d) - and x
». 9 + 5 + 4 x 3 - 6 * 1 2
   (a) + and x *(b)* + and x (c) + and - (d)+ and -
10. 12 + 2 x 3 + 8 5 16
   (a) + and + (6) and + (c) x and + (d) + and x
11. Which of the following two signs need to be interchanged to make the given equation correct ? (M.B.A. 1997)
   10 + 10[V]+ 1 0 - 10 x 10 = 10
   (a) + and- (6) + and+ (c)+andx (d) + and

Directions (***Questions 12 to 16) : In each of the following questions, the two expressions on either side of the sign (=) will have the same vtilue if two terms on either side or on the same side cm interchanged. The correct terms to be interchanged have been given as one of the four alternatives under the expressions. Find the correct alternative*** in ***each case.*** (CA.T. 1997)

12. 5 + 3 x 6 - 4 + 2 = 4 x 3 - 1 0 + 2 + 7
    (a) 4, 7 (6) .5. T (c) 6, 4 *id)* 6, 10
13. 7 x 2 - 3 + 8 + 4 = 5 + 6 x 2 - 2 4 + 3
    (a) 2. 6 (6) 6, 5 (c) 3, 24 *(d)* 7, 6
14. 15+ 3 x 4 - 8 + 2 = 8 x 5 +16+ 2 - 1
    (a) 3, 5 (6) 15, 5 (c) 15, 16 (rf) 3, 1
15. 6 x 3 + 8 + 2 - 1 = 9 - 8 + 4 + 5 x 2
    (a) 3, 4 (6) 3, 5 (c) 6, 9 (d) 9. 5
16. 8 + 2x 5 - 11 +9 - 6 x 2 - 5 + 4 + 2
    (a) 5, 9 (6) 8. 5 (c) 9, 6 (d) 11, 5

Directions (***Questions 17 to 20) : In each of the following questions, which one of the four interchanges in signs and numbers would make the given equation correct f***

17. 6 x 4 + 2 = 1 6
    (a) + and x, 2 and 4 (6) + and x, 2 and 6
    (c) + and x, 4 and 6 (d) None of these
18. (3+ 4)+ 2 = 2
    (a) + and +, 2 and 3 *(b)* + and +, 2 and 4
    (c) + and +, 3 and 4 (d) No interchange, 3 and 4
19. 4 x 6 - 2 - 14
    (a) x to +, 2 and 4 (6) - to +, 2 and 6
    (c) - to +, 2 and 6 (d) x to +, 4 and 6
20. <6 +2) x 3 - 0
    (a) + and x, 2 and 3 (6) x to 2 and 6
    (c) + and x, 2 and 6 *id)* x to 2 and 3

ANSWERS /

1. (c): On interchanging - and + and 4 and 8 in (c), we get the equation as 8 - 4 + 2 - 6 or 8 - 2 - 6 or 6 - 6. which is true.
2. (c): On interchanging + and * and 4 and 5 in (c). we get the equation as 4 + 5 x 2 0 = 104 or 104 = 104. which is true.
3. (6): On interchanging + and - and 4 and 8 in (6), we get the ec^iation as 8 4 - 12 = 0 or 12 - 12 = -ft or 0 - 0 , which is true.
4. *(b)*: On interchanging - and * and 3 and 6 in (6), we get the equation as 6 x 3 - 8 = 10 or 1 8 - 8 - 1 0 or 10 - 10, which ia true.
5. (6): On interchanging - and +, we get the equation as

   2
   5 + 3* , 8 + 1 2 - 4 = 3 or 5 + 3 x ~ - 4 - 3 or 3 - 3 , which is true.
   o
6. (a): On interchanging + and *, we get :
   Given expression = 5 * 6 * 3 ~ 1 2 ^ 2 - 5 + 6 x 3 - 6 « 5 + 1 8 - 6 = 1 7 .

7. (a): On interchanging x and +, we get :
Given expression = 2 + 3 x 6 - 12 + 4 = 2 + 3 x 6 - 3 = 2 + 18 - 3 - 17.
8. *(b):* On interchanging - and +, we get :
Given expression = 16 + 8 - 4 + 5 x 2 = 2 - 4 + 5 x 2 - 2 - 4 + 10 = 8,
9. (c): On interchanging + and we get :
Given expression «9 + 5 - 4 x 3 + 6 = 9 + 5 - 4 x | = 9 + 5 - 2 = 12.

10. (6): On interchanging - and +$_f$ we get :
Given expression = 12 + 2 + 6 x 3 - 8 - 6 + 6 x 3 - 8 = 6 + 18 - 8 = 16.
11. (c): On interchanging + and x$_f$ we get the equation as
10 x 10 + 10 - 10 + 10 = 10 or 10 x 1 - 10 + 10 = 10 or 10 = 10, which is true.
12. (c): On interchanging 6 and 4 on L.H.S., we get the statement as
5 + 3 x 4 - 6 + 2 = 4 x 3 - 10 + 2 + 7 or 5 + 12 - 3 = 12 - 5 + 7 or 14 = 14, which is true.
13. (d): On interchanging 7 and 6, we get the statement as
6 * 2 - 3 + 8 + 4 = 5 + 7 x 2 - 24 + 3 or 12 - 3 + 2 - 5 + 14 - 8 or 11-11, which is true.
14. (o): On interchanging 3 and 5, we get the statement as
15 + 5 x 4 - 8 + 2 = 8 x 3 + 16 + 2 - 1 or 15 + 20 - 4 = 24 + 8 - 1 or 31 = 31, which is true.
15. (d): On interchanging 9 and 5 on R.H.S., we get the statement as
6 x 3 + 8 + 2 - 1 = 5 - 8 + ^ + 9 x 2 or 18 + 4 - 1 - 5 - 2 + 18 or 21-91. which is true.
16. (c): On interchanging 9 and 6, we get the statement as
8 + 2 x 5 - 11 + 6 - 9 x 2 - 5 + 4 + 2 or 4 x 5 - 11 + 6 = 18 - 5 + 2 or 15 = 15, which is true.
17. (c): On interchanging + and x and 4 and 6, we get the equation as
4 + 6 x 2 = 16 or 4 + 12 = 16 or 16 = 16, which is true.
18. *(a):* On interchanging + and + and 2 and 3, we get the equation as
(2 + 4) • 3 - 2 or 6 + 3 - 2 or 2 - 2, which is true.
19. (c): On changing - to + and interchanging 2 and 6, we get the equation as
4 x 2 + 6 = 14 or 8 + 6 = 14 or 14 = 14, which is true.
20. (d): On changing x to - and interchanging 2 and 3, we get the equation as
(6 + 3) - 2 - 0 or 2 - 2 - 0 or 0 - 0, which is true.

TYPE 3 : DERIVING THE APPROPRIATE CONCLUSIONS

Ex. 1. It being given that x denotes 'greater than', $ denotes 'equal to\ < denotes 'not less than[1], 1 denotes 'not equal to\ A denotes 'less than* and + denotes 'not greater than*, ' (M.B.A. 1998)
choose the correct statement from the following :
If *a* x *b* A c, it follows that

(a) a $ c A *b* (6) *b* < a x c (c) a < 6 + c
(d) c + b < a *(e) b < a $ c*

Sol. Using the usual notations, we have :

(a) : The statement is a > 6 < c => a « c < 6, which is false. (%« c > 61
(6) : The statement is a > 6 < c => 6 4 a > c, which is false. [•.• 6 < a)
ug (c) : The statement i s a > 6 < c = * a < t 6 i c , which is true.
(d) : The statement is a > 6 < c c * 6 4 a, which is false. [•.• 6 < *a]*
(e) : The statement is a > 6 < c o 6 < t a = c, which is false. [•.. *b* < *a]*

Hence, the statement (c) is true.

Ex. 2. In the following questions, the symbols @ and @ are used with the following meanings : (S.B.I.P.O. 1997)

'A * B' means 'A is greater than B';

'A * B' means 'A is either greater than or equal to B';

'A = B' means 'A is equal to B';

'A @ B' means 'A is smaller than B';

'A @ B' means 'A is either smaller than or equal to B\ *

Now, in each of the following questions, assuming the given statements to be true, find which of the two conclusions I and II given below them is/are definitely true ?

Give answer (a) if only conclusion I is true; (fc) if only conclusion II is true; (c) if either I or II is true; (*d*) if neither I nor II is true and *(e)* if both I and II are true.

1. Statements : M = T, T @ Z, S • M
   Conclusions : I. Z • M   II. Z = M
2. Statements : R @ M, M • P, R • L
   Conclusions : I. M = L   II. P = L
3. Statements : L@C, C • Z, Z @ F
   Conclusions : I. C • F   II. F = C
4. Statements : Z @ B, N • S, B @ N
   Conclusions : I. B = Z   II. S @ B
5. Statements : T *P, P @ S, P = M
   Conclusions : I. S • M   II. T @ S

Sol. 1. Given statements : M = T, T S Z, S > M

Now, to verify conclusions I and II, we need to find a relation between Z and M.

Z £ T, T = M   Z £ M

=> Z > M or Z = M i.e., Z • M or Z = M.

So, either I or II follows.

Hence, the answer is (c).

2. Given statements : R £ M, M > P, R > L

I. Relation between M and L :

M > R, R > L   M > L i.e., M • L.

So, I is not true.

II. Relation between P and L.

P < M, M 2 R, R £ L => no definite conclusion.

So, II is also not true.

Hence, the answer is *(d).*

3. Given statements : L < C, C > Z, Z £ F.

Clearly, we find a relation between C and F.

C > Z, Z £ F => no definite conclusion.

So, neither I nor II is true.

Hence, the answer is (d).

4. Given statements : Z < B, N 2 S, B < N.   v ,

I. Relation between B and Z :

Clearly, B > Z i.e., B • Z.

So, I is not true.

II. Relation between S and B :
S £ N, N > B no definite conclusion.
So, II is also not true.
Hence, the answer is (d).

5, Given statements : T * P, P < S, P = M
I. Relation between S and M :
S > P, P = M => S > M i.?., S • M.
So, I is true.
II. Relation between T and S :
T > P, P < S => no definite conclusion.
So, II is not true.
Hence, the answer is (a).

EXERCISE 12C

1. Which of the following conclusions is correct according to the given expressions and symbols ? *(VD.C.* 18*5)
A : * B : > C : * E : 4 F : <
Expressions : (aEb) and (bEc)
(a) aEc (6) aFc (c) cBa 1 (d) cBb

2. Find the correct inference according to given premises and symbols :
A : Not greater than B : Greater than C : Not equaHo
D : Equal to E : Not less than F : Less than
Premises : (/Cm) and (/Am)
(a)lhrn - (6)/Dm (c)/Em (d) /Fm
(Transmission Executives', 1094)

Directions (*Questions 3 to 8) : It being given that :*
A denotes 4equal to9; • denotes 4not equal to9; + denotes \greater than9;
- denotes 4 less than9; x *denotes 4not greater than9; + denotes 'not less than9.*
Choose the correct statement in each of the following questions :

3. *a - b* - c implies
(a) a-b + c (6) 6 + a - c (c) c x 6 • a <d) 6 + a • c
a + b -c implies i
(a) 6 - c - a (6) c - 6 + cj (c) c + 6 - o (d) c x 6
a x 6 + c implies
(a) a - 6 + *c* (6) c x 6 «• a (c) a • b • c (d) 6 + a + c
a + 6 • c does not imply
(a) 6 - a + c (6) c - *b* - a (c) c - a + 6 (d) 6 - a - c
a 4 *b* - c does not imply
(a)c + 6 - a (6) 6 - a + c (c) b • a • c (d> None of these
a • *b* • c implies
(a) a + *c* (6) a - 6 - c (c) *a* + 6 + c (d) None of these

Directions (*Questions 9-10) : If* a *means 4greater than9,* p *means 4equal to* 0 *means 4not less than\ y means 4less than\ 8 means 4not equal to9 and*

H means <not greater than\ then which of the four alternatives could be a correct or proper inference in each of the following f (P.C.S. 1995)

9. *a a 2b* and *2b* 0 r
   (a) a n r (b)*aar* (c)a(Jr (d)*ayr*
10. 2x6 *y* and y P 3*
    (a)y8 6x , (6) 2x n 3* (c)2r8 3*z* (d) 3z n 3y
11. If A stands for 'not equal to' <*), B stands for 'greater than' (>), C stands for 'not less than* (<t), D stands for 'equal to' (=), E stands for 'not greater than' (i), F stands for 'less than* (<), then according to the given premises (4x F *by)* and (5y E 3s), which of the following inferences is correct ? (C.B.I. 1994)
    (a) 4x A 3s <6>4xB3s (c)4*C3s <*d)ixD3s*

Directions (*Questions 12 to 17) : In the following questions%*

A *means 4is greater than\ % means 'is lesser than* • *means lis equal to9 = means *x* not equal to\ + means 4is a little more than9,* x *means 4is a little less than9.*

Choose the correct alternative in each of the following questions.

12. If a A 6 and 6 + c, then
    (a) a %c *(b)c%a* (c) c + *a* (d) Can't say
13. If c • *a* and *a* = *b*. then
    (a) 6 A a (6) c • *a* (c) *b* - a *{d)* Can't say
14. If *a* x *b* and 6 • c, then
    [a)c-a (6) 6 Ac (c)a+c (d) c • a
15. If *c % b* and *b* x a, then
    (a) a Ac (6) c • a (c)6Dc (d)cAa
16. If *ac* • 6c, then • >
    (a) a • c (6) 6 Ac <c)cA6 *(d)b%a*
17. If *ac % bd* and *ab* A cd, then
    (a) 6 • c (6) 6 A a *(c) a %c* (d) Can't say

Directions (*Questions 18 to 22) : In each of the following questions, the Greek letters standing for arithmetical operations are given. Find the relationship which can'definitely be deduced from the two relationships given at the top.*

Operations : a is 'greater than', p is 'less than', y is 'not greater than', 8 is 'not less than', 0 is 'equal to'.

18. If A a 2C and 2A 9 3B. then
    (a) C p B (6) C 5 B (c) C a B (d) COB
19. If 3A a B and 3B a 2C, then
    (a) 4AaC <6)5AaC (c)2A0C (d)3A8C
20. If B0 2C and 3Cy A, then
    (a) B 8 2A (6) B G A (c) 3B a 2A <d) B p A
21. If 3C8 2A and BaC, then
    (a) 2Aa 3B (6) 3B a 2A <c)B6A <d) 3B 0 2A
22. If 3B 0 2C and 2A a 3C. then
    (a) B 8 A (6) BOA (OBPA (d)BaA

Directions (*Questions 23 to 27) : In the following questions the symbols ©, ©, @ and ~ are used with the following meaning :*

*© means 'greater than'; © means 'either greater than or equal to'; @ means *smaller than*; @ means ⁴either smaller than or equal to*; - means 'equal to'.*

Now in each of the following questions, assuming the given statements to be true, find which ofthe two conclusions I and II given below them is/are definitely true ?

Give answer (a) if only conclusion I is true; (6) *if only conclusion II is true; (e) if either I or II is true, (d) if neither I nor II is true and (e) if both I and II are true.*

23. Statements : M « N, L © N, M = P

Conclusions : I. N = P　　II. N « P

24. Statements : A ® C, M © F, C © F

Conclusions : I. M = A　　II. C © M

26. Statements : B © P, C © N. P = N

Conclusions : I. P @ C　　II. C © B

26. Statements : K@P, Z © K, K © M

Conclusions : I. Z = M　　II. Z © M

27. Statements : Z@P, T = M, M © Z

Conclusions : I. M © Z　　II. T © P

Directions (*Questions 28 to 32) : In the following questions, the symbols, ©, ©, =, » and * are used with the following meanings :* (Bank P.O. 1997)

%P © Q ' means 'P is greater than Q';

'P © Q ' means 'P is greater than or equal to Q';

'P-Q' means 'P is equal to Q';

'P • Q ' means 'P ie smaller than Q';

'P$Q' means *P *is either smaller than or equal to Q*

Now in each of the following questions, assuming the given statements to be true, find which ofthe two conclusions I and II given below them is/are definitely true.

Give answer (a) *if only conclusion I is true;* (6) *if only conclusion II is true;* (c) *if either I or II is true; (d) if neither I nor II is true and* (e) *if both I and II are true.*

28. Statements : P © T, M * K, T = K

Conclusions : I. T © M　　II. T = M

29. Statements : S • M, M © L, L © Z

Conclusions : I. 8 = Z　　II. S * L

30. Statements : D © F, F = S, S * M

Conclusions : I. D © M　　II. F © M

31. Statements : J - V, V • N, R • J

Conclusions : I. R • N　　II. J © N

32. Statements : L © U, C • L, C © B

Conclusions : I. U = C　　II. L © B

Directions (***Questions 33 to 35) : In the following questions :***

'P • Q * means 'P is greater than Q';

'P *Q¹ means 'P is either greater than or equal to Q';

'P = Q' means 'P is equal to Q';

'P • Q' means 'P is smaller than Q';

'P • Q' means *P is either smaller than or equal to Q

In each question, a statement is given followed by two conclusions I and II. You are to consider each statement and the conclusions that follow and decide which of the conclusions is/are implicit ? , (Assistant Grade, 1998)

33. Statements : G • S, **F •** S, **T •** G. /

Conclusions : I. F • T **II. T** = S.

I —•

(a) Both **I** and **II** are implicit *(b)* Only **I** is implicit

(c) Neither **I** nor **II** is implicit *(d)* Only **II** is implicit

34. Statements : M = N, N • B, B • P

Conclusions : I. P = N **II. B • M**

(a) Only **I** is implicit (6) Only **II** is implicit

(c) Both **I** and **II** are implicit *(d)* Neither **I** nor **II** is implicit

35. Statements : N • T , T = P

Conclusions : I. P • N **II. P = N**

(a) Either **I** or **II** is implicit (b) Only **I** is implicit

(c) Only **II** is implicit * *id)* Neither **I** nor **II** is implicit

Directions *(Questions 36 to 39) : Assume the following :*

'A® B' means 'A is greater than B';

⁴A • B' means *A is either greater than or equal to B';

'A$ B* means 'A is equal to B';

'A* B' means 'A is smaller than B';

'A # B' means *A is either smaller than or equal to B

In each question, two statements followed by two conclusions I and II are given. Assuming the statements to be true, state which of the conclusions I and II is/are definitely true f (M.B.A. 1998)

Give answer (a) if only conclusion I is true; (6) if only conclusion II is true; (c) if either I or II is true; (d) if neither I nor II is true; and (e) if both I and II are true.

36. Statements : P * Q, M • N **$ P**

Conclusions : 1. M @ P **II.** N **# Q**

37. Statements : L • M, R • T $ L

Conclusions : I. T • M **II. R @ L**

38. Statements : X @ Y @ Z , U @ Z $ V

Conclusions : I. V * U **II. X @ V**

39. Statements : G *** H # K , H @ Q $ R**

Conclusions : I. G **$ Q** **II. R •** G

Directions *(Questions 40 to 44) : In the following questions,* a ***stands for 'equal to';*** JJ ***for "greater than¹;*** y ***for 'less than' and*** 8 ***for 'not equal to'.***

j (Hotel Management, 1996)

40. If 6x a 5y and 2y p 3z, then
(a) 2* p 3z (fc) 4x p 3z (c) 2x y z *(d) ix* a 3z

41. If a* y 6y, 6x a cz and 6^2 a ac, then
(a) ax pcy (fc)ayctcz (c)yyz (d)ypz

42. If a6xy a c^2z, *b\$ay* and *b^2 a ac,* then
(a) ax^2 p *cz* *(b) a^2x^2* p *cz* (c) *b^2x* p *c^2z* *(d) bx^2* p *c^2z*

43. If *bey* y *ax,cyabz* and a^2 y 6c, then
(a)cxaafez (6)cxya6z *(c)cx6abz* *(d)c^2xya^2z*

44. If a?r a *byz, czx* a 6^2y and c^2z a ary, then
(a)a6caxyz (6)a6cpxyz *(c)abcbxyz* *(d)abcyxyz*

45. If A + B > C + D, B + E = 2C and C + D > B + E, it necessarily follows that :
(a) A + B >2C (fe)A+B>2D (c)A>B>2E *(d)* A > C

(Hotel Management, 1995)

46. If A + D > C + E, C + D = 2B and B + E > C + D, it necessarily follows that
(a) A + B > 2D (6) B + D > C + E (c)A + D > B + E (d)A + D > B + C

(Hotel Management, 1995)

Directions (Questions 47 to SI) : In each of the questions given below, use the following notations : **(S.B.1.P.O. 1997)**

A B means 'add B to A'; ***A B means \subtract B from A';***
A ® B means divide A by B'; ***A* B means *multiply A by B***

Now, answer the following questions.

47. The time taken by two running trains in crossing each other is calculated by dividing the sum of the lengths of two trains by the total speed of the two trains. If the length of the first train is Li, the length of the second train is L2; the speed of the first train is Vi and the speed of the second train is **V2.** which of the following expressions would represent the time taken ?
(a) (Li" U) • (V f V_2) (6) (Li" L2) ® (Vi" V_2)
(c) **KLI" L2> @ (VR** V_2)i **• 60** **(<F) (LR** L_2**> @ (VR** V_2)
(e) None of these

48. The total airfare is calculated by adding 15% of basic fare as fuel surcharge, *2%* of the basic fare as LATA charges and Rs 200 as airport tax to the basic fare. If the basic fare of a sector is B, which of the following will represent the total fare ?
(a) B" (B • 15) @ 100" (B • 2) @ 200" 100
(б) B" (B • 15) @ 100" (B * 2) @ 100" 200
(c) B" (B • 15) @ 100' (B • 2) @ 100" 200
(d) B' (B • 15) @ 100" (B • 2) @ 100" 200
(e) None of these

49. The profit percentage of a commodity is worked out by multiplying the quotient of the difference between the amount of sale price and the total expenses and divided by the
amount of total expenses by 100. If the sale price of an article is S, the total expenses are equal to the sum of the cost price (C), transportation

costs (T), labour charges (L), which of the following expressions would indicate the profit percentage ?

(a) [{S - (C • L + T)} + (C + L + T) x 100) (6) US' (C"L" T)> 0 (C"L" T) @ 1001

(c) C(S# (C"L" T)} ® (C"L" T) • 100] (d) US" (C'L' T)} • (C"L" T) 0 100)

(e) None of these

60. While considering employees for promotion, an organisation gives 2 marks for every year of service beyond the first two years, four-thirds of the marks obtained in an examination out of 90 marks, five marks for each level of education-matriculation, graduation and post-graduation. Which of the following represents the total marks a candidate gets if he has put in T years of service, obtained K marks in the examination and passed Xth, XIIth and Graduation level examinations ?

(a) (T2) • 3" 5 • 2 " 4 • T 0 3 (6) (K'2) • 2" 5 • 3 " 4 • T © 3

(c) (T'2) » 2" 5 * 3 " 4 » K 9 3 *(d)* (T2) • 2" 5 • 3 " 4 • K @ 3

(e) None of these

51. In a semester system of examination, the total marks obtained is arrived at by adding 10% of the marks obtained in first periodical, 15% of the marks obtained in the second periodical and 75% of the marks obtained in the final examination. If a student secures P marks out of 150 in first periodical, T marks out of 180 in second periodical and M marks out of 400 in the final examination, which of the following will represent the total marks obtained by him ?

(a) (P © 150 • 10)" (T@400 • 15)" (M @ 180 • 75)

(h) (P @ 150 • 10)" (T ® 180 • 15)" (M « 400 * 75)

(c) <P • 150 • 10)" (T • 180 « 15)" (M • 400 ® 75)

(d) (P @ 10 • 10)" (T % 180 • 15)" (M 8 400 • 75)

(e) None of these

ANSWERS

1. (a): *aEb* and 6Ec *at b* and *blc=*alc=> aEc.*

2. *id)*: /Cm and /Am ^ *H m* and *li m I <m IFm.*

S. (6): With usual notations, we have :

(a) a <b <c => a < 6 > c, which is false.

ib) a <b <c => 6 >*a* < c, which is true.

(c) a <b<c ^ c i *b>a,* which is false.

(d) a <b<c b>a i c, which is false.

4. (c): With usual notations, we have :

(a) o > 6 < c 6 < c < a, which is false

(б) a > 6 < c => c < 6 > a , which is false,

(c) a > fc < c c > 6 < a , which is true.

id) a >b<c c ^ & * a, which is false.

5. (b): With usual notations, we have :

(a) a * *b A c => a <b>c,* which is not true.

ib) a 1 b k c => cl b 4Ta. which is true,

(c) a * *b* c a *b*c,* which is not true.

ii c ^ M ai c, which is not true.

6. *(d):* With usual notations, we have :

(a) a > b > c b<a>c, which is false.

(6) a > 6 > c c < 6 < a, which is false
(c) a > 6 > c i> c < a > 6. which is false.
(</) a > 6 > c i> 6 < a < c, which is true.

7. (6): With usual notation*, wc have :
(o) a > 6 < c 4> c > 6 < a, which is false.
(6) a > 6 < c *4> b* < a > c, which is true.
(c) a > 6 < c 6 * a * c, which is false.

8. (d): With usual notation*, we have :
*(a) a * b * c => a >b>c,* which is false.
(б) a * 6 * c *a <b <c9* which ia false.
*(c) a * h * c a i b t* c. which is false.

9. (6): (a a 26) and *(2b* 0 r) =» a > 26 and 26 * *r*
a > 26 and 26 * r : *a >r i£. a a* r.

10. <c): (2* 6y) and (y P 3/) 2* * y and y a 3*
o 2x * 3* i^.. 2* 5 3*.

11. (a): (4* F 5 » and (5y E 35) (4r < 5y) and (5y i 3t)
(4x < 5y) and (fiy £ 3s)
Ax < 3s or *4x* * 3*
4x F 3* or 4r A 3s.

12. (6): a A 6 and 6 • c => a > 6 and 6 is a little more than c.
=> A > c = ^ c **< A (J**. c % a.

13. (c): c = o and a = 6 **DCHO** and a * 6 => *b *a i*.b = a.*

14. (a): a x 6 and 6 • c a is a little less than 6 and 6 «c
=> o is a little less than c
=> c is a litUe more than'a le.c + a

15. (a): c ft 6 and 6 * a => *c<b* and 6 is a little less than a.
=> *c<a a>c t.e.* a Ac.

16. (</): ac • 6c *ac>bc* => a > 6 => 6 < a *ie. b%a*

17. (d)*: ac% bd* and *ab Acd ac <bd* and a6 > erf.
Clearly, no conclusion can be drawn.

18. *(a) :* A a 2C and 2A 9 3B ^ A > 2C and 2A = 3B
2A > 4C and 2A = 3B
=> 3B > 4C C < B i.e. C P B.

19.(6): 3Aa B and 3Ba2C 3A > B and 3B > 2C
=> 3A > B and | B > C
^ | A > | B and | B > C
| A > C 5A > C Lt. 5A a C.

20. (rf): B 0 2C and 3C y A B * 2C and 3C ^ A
B =2C and 3C s A
B - 2C 3C i A
B < A *i.e.* B p A

21.(6): 3C 6 2A and B a C 3C* 2A and B > C
3C > 2A and B > C '
3B > 3C and 3C 2 2A
3B > 2A *i.e.* 3B a 2A.

22L ic): 3B 9 2C and 2A a 3C 3B = 2C and 2A > 3C.

| B = 3C and 3C < 2A

| B<2A B < | A

=> B < A i.e., Bp A

23. *id)*: Given statements : M £ N, L > N, M = P.

To verify the given conclusions, we find a relation between N and P.

Now, N £ M, M * P 2 N £ P.

Clearly, both I and II are false

24. *ib)*: Given statements : A S C, M s F, C > F

I. Relation between M and A :

M * F, F < C, C ^ A no definite conclusion.

So, I is not true.

II. Relation between C and M :

C > F, F * M => C > M *Le.* C©M.

So, II is true.

25. *ie):* Given statements : B £ P, C > N, P = N

I. Relation between P and C :

P = N. N < C => P<C i.e. P<C.

So, I is true.

II. Relation between C and B :

C > N, N = P. P 2 B => C > B i.e. C © B.

So. II is true.

26. (6): Given statements : K < P, Z > K. K * M

Relation between Z and M :

Z> K, K * M Z > M *i.e.* Z©M.

So, I is false and II is true.

27. *ie):* Given statements : Z < P, T = M. M £ P

I. Relation between M and Z :

M * P, P > Z M > Z *i.e.* M © Z.

So, I is true.

II. Relation between T and P.

T = M, M £ P => T * P *ie.* T©P

So, II is true.

28. ic): Given statements : P > Q, M £ K, T = K.

Relation between T and M :

T = IC M T ^ M T >M orT» M

=> T ® M or T = M.

So, either I or II is true.

29. *id):* Given statements : S < M, M > L, L 2 Z

I. Relation between S and Z :

S < M, M > L, L £ Z => no definite conclusion.

So, I is not true.

II. Relation between S and L :

S < M, M > L => no definite conclusion.

So, II is also not true.

30. (d): Given statements : D > F, F = S, S £ M
   I. Relation between D and M :
      D > F, F = S, S < M => no definite conclusion.
      So, I is not true.
   II. Relation between F and M :
      F - S . S s M =» F s M.
      So, F © M *i.e.* F i M is not true.
      Thus, II is false.
31. (a): Given statements : J = V, V < N, R s J
   I. Relation between R and N :
      R s J, J « V, V < N =»' R < N ie R * N.
      So, I is true.
   II. Relation betwoen J and N :
      J = V, V < N => J < N *Le.* J • N.
      So. J © N *Le.,* J i N is not true.
      Thus, II is false.
32. (6): Given statements C < L , C > B
   I. Relation between U and C :
      U s L, L > C => no definite conclusion.
      So, I is not true.
   II. Relation between L and B :
      L > C, C > B => L > B U. L©B.
      So, II is true.
33. (6): Given statements : G £ S, F i S, T < G
   I. Relation between F and T :
      F * S, S £ G, G > T =» F > T U. F • T
      So, I is true.
   II. Relation between T and S :
      T < G, G s S => T < S U. T • S.
      So, T » S is not true.
      Thus, II is false.
34. (b): Given statements : M = N, N > B, B < P
   I. Relation between P and N :
      P > B, B < N =» no definite conclusion.
      So, I is not true.
   II. Relation between B and M :
      B < N, N = M =» B < M *Le.* B • M.
      So, II is true. i
35. (a): Given statements : N ST, T- P.
   Relation between P and N :
   P - T, TJsN => P i N => P > N or P• N
   => P • N or P = N.
   So, either I or II is implicit.
36. *(b):* Given statements : P s Q, M £N = P
   I. Relation between M and P :
      M 2 N = P M 2 P ie. M • P.
      So, I is not true.

II. Relation between N and Q :
N = P, P s Q => N s Q *i.e.* N # Q.
So, II i^true.

37. (a): Given statements : L * M , R * T = L
I. Relation between T and M :
T - L , L * M => TfcM i.e. T • M.
So. I is true.
II. Relation between R and L :
R ^ T - L R * L i.e. R • L.
So, II is not true.

38. *ie)*: Given statements : X > Y > Z , U > Z - V
I. Relation between V and U :
V - Z < U => V < U U. V * U.
So, I is true.
II. Relation between X and V :
X > Y > Z => X > Z.
Now, X > Z and Z = V » X > V *i.e.* X ® V.
So. II is true.

39. *id):* Given statements : G < H £ K, H > Q = R.
I. Relation between G and Q.
G < H, H > Q ^ no definite conclusion.
So, I is not true.
II. Relation between R and G :
R * Q < H R< H.
Now, R < H and H > G no definite conclusion.
So, II is not true.

40. *ib)*: 6x a 5y and 2y p 3* 6* = 5*y* and 2*y* > 3*z*

=> 6* » 5y and *y* >

=> *6x m5y* and 5y > ISZ — => *Sx* > —

=> *I2x* > 15* => 4x > &
4r>3z U. 4xp3*.

41. (d): *axyby%bxacz* and b^2 *a ac* *ax<by%bx=cz* and 6^2 = ac.
bx =*cz* => 6^2x = *bcz* ==> *acx* = 6cz => *ax* = *bz.*
ax<by => *bz* < *by* => 2 < y y > z i.e. y p

42. (a): *abxy a c z, bx $ ay* and 6 aac » *abxy* = *c z9bx* > *ay, b* = ac.
Now, *bx* > *ay* => > *aby*

acx > a6y (•.• 6^2 = ac)

» *cx>by* *by* < *ex.*

c^2z » a6xy » axfty < *axcx* *cz* < ax^2

=> ax^2 > *cz* *i.e.*

cz.

43. (c): *bey* y ax, cy a *bz* and a^2ybc => 6cy < ax, cy • 6*. a^2 < 6c
*cy = 6** c^2y - *bcz* >a^2z => c^2y>a^lz.
ax > *bey* > a^2v ax > a^2y x > ay

$\Rightarrow$ *cx> acy* $\Rightarrow$ *cx > abz* *(•.• ey • bz)*

$\Rightarrow$ *cx * abz ijt. cxbabz.*

44. (a): *a^2x a byz, czx a b^2y* and *c^2z* a *axy* => *a^2x * byz, czx - b^2y, c^2z = axy.*

czx = b^2y *c^2zx « cb^2y* *axyx-cb^2y* *(-.c^2z *axy)*

=> $ax^2 = cb^2$.

Now, *a^2x = byz* => *a^2x^2 = bxyz* *a.ax^2 ~ bxyz*

=> *acb^2 - bxyz* (v ar^2 = $c6^2$)

=> *abc - xyz ia. abc* a *xyz.*

45. (a): A + B > C + D. C + D > B + E. B + E = 2C

=> A • B > B E, B E = 2C => A + B > 2C.

46. (</): A D > C + E =» A + D > (2B - D) + E (v C + D - 2B)

A + D > (B + E) + (B - D)

» A + D > (C + D) + (B - D)

=>' A + D > B + C.

47. (6): Clearly, time taken $\frac{\text{sum of lengths of two train*}}{\text{total speed of two trains}}$

$\frac{Li + L2}{V| \cdot Va}$ (Li"U> ® (Vi"V_2).

48. *(b)*: Total fare B + 15%ofB + 2%ofB + 200

= B + $\frac{B * 15}{100}$ + $\frac{B * 2}{100}$ 2Q()

a B" (B • 15) @> 100" (B • 2)® 100"200.

49. (c): Profit percentage = + * 100

= (S' (C"L^M ***T))*** ® (C"L" T) • 100

50. (e): Clearly, total marks = (T - 2) * 2 + $\frac{4K}{}$ + 5 * 2

= (T2) • 2"4 • K 0 3"6 • 2.

51. (6): Marks out of 150 in first periodical • P.

Marks out of 100 in first periodical = P^ x 100 I

Marks out of 180 in second periodical = T.

Marks out of 100 in second periodical i i o $^{x\ 1\ 0\ 0}$) '

Marks out of 400 in final examination = **M**

Marks out of 100 in final examination

Total marks

1 0 % o f ^ . 1 0 o J . [1 5 % o f ^ . 1 0 0 | . [7 5 % . f ^ X 100

M
400

(P ® 150 • 10)" (T® 180 • 15)" (M @ 400 • 75).